

OLIVER AGDM, AB, Canada

WMO: 713510

Lat: 53.6466N

Lon: 113.3546W

Elev: 2182

StdP: 13.57

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-26.0	-18.8	-32.2	1.0	-25.8	-24.8	1.5	-18.4	18.8	27.0	16.5	26.7	2.2	340	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	83.5	64.0	80.2	62.3	77.3	60.9	66.9	79.4	64.8	76.3	62.8	73.8	5.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	91.5	71.9	60.3	84.6	69.6	58.3	78.9	67.5	32.7	79.3	31.0	76.7	29.4	73.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	14.2	12.0	DB	-35.7	89.7	7.3	3.6	-40.9	92.3	-45.2	94.5	-49.3	96.5	-54.6	99.2	(4)
(5)			WB	-35.9	70.7	7.3	2.2	-41.1	72.3	-45.4	73.6	-49.5	74.8	-54.7	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	13.5	13.7	24.8	39.3	51.3	58.7	63.0	60.2	51.2	39.6	24.2	12.6		
	DBStd	21.43	15.88	14.95	13.70	9.48	7.96	5.27	5.06	5.70	7.89	8.50	13.03	14.64		
	HDD50	5695	1132	1016	780	337	79	4	0	2	78	335	773	1159		
	HDD65	9994	1597	1436	1245	772	427	199	100	166	417	788	1223	1624		
	CDD50	1244	0	0	0	14	120	264	402	319	113	12	0	0		
	CDD65	71	0	0	0	0	3	9	37	19	3	0	0	0		
	CDH74	1482	0	0	0	3	118	207	613	391	141	9	0	0		
	CDH80	318	0	0	0	0	18	29	151	84	36	0	0	0		
	WSAvg	5.0	4.6	4.5	5.0	6.2	6.0	5.3	4.7	4.4	4.8	5.2	4.6	4.3		
	PrecAvg	16.5	0.7	0.5	0.6	1.1	1.9	2.7	3.5	2.1	1.1	0.8	0.6	0.7		
(15) Precipitation	PrecMax	21.8	3.3	1.2	1.9	1.9	3.9	5.7	5.7	4.5	3.0	2.1	1.3	1.3		
	PrecMin	9.3	0.1	0.2	0.2	0.1	0.3	0.8	0.7	0.7	0.0	0.2	0.1	0.2		
	PrecStd	3.1	0.7	0.2	0.4	0.5	0.9	1.2	1.2	1.0	0.7	0.5	0.3	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	48.7	56.3	73.5	82.5	83.7	88.9	86.8	85.1	74.1	55.8	45.8		
		MCWB	38.1	39.2	43.7	52.0	57.3	62.6	67.6	65.1	61.7	54.6	44.2	38.1		
	2%	DB	42.6	43.0	50.6	66.9	77.4	79.5	84.0	81.9	77.7	63.9	48.5	41.2		
		MCWB	36.3	35.7	40.2	48.4	54.8	60.4	66.2	64.0	58.5	50.2	40.1	35.3		
	5%	DB	39.5	39.0	46.3	60.5	72.9	75.9	80.4	78.3	72.4	58.4	43.8	37.7		
		MCWB	34.2	33.2	38.3	45.4	53.2	58.8	64.5	62.8	57.0	46.5	37.3	32.7		
	10%	DB	35.7	35.4	42.3	55.9	68.5	72.3	77.0	74.9	66.8	53.9	40.2	33.8		
		MCWB	31.7	31.1	35.7	43.0	51.2	57.4	62.8	61.4	54.1	44.1	35.2	29.8		
	0.4%	WB	38.9	39.7	44.1	52.3	59.1	65.8	71.0	68.2	63.0	55.4	45.2	38.9		
		MCDB	45.0	47.8	53.7	71.8	76.3	78.8	82.4	80.6	79.6	72.5	53.3	44.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.6	35.9	40.9	49.1	56.4	62.7	68.1	66.0	60.1	50.9	40.9	35.6		
		MCDB	42.0	42.0	49.7	64.8	71.8	74.4	80.9	78.7	75.4	62.7	47.1	40.3		
	5%	WB	34.6	33.6	38.5	46.2	54.6	60.7	65.9	64.0	57.7	47.5	38.1	33.0		
		MCDB	39.1	38.4	45.8	58.7	69.7	72.2	77.5	75.4	70.4	56.4	42.9	37.5		
	10%	WB	31.8	31.2	36.0	43.7	52.6	58.9	63.7	62.1	55.3	45.0	35.6	29.8		
		MCDB	36.0	35.4	41.5	54.4	66.4	69.7	74.4	73.2	65.0	52.4	39.7	33.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.2	22.5	21.1	21.5	24.6	21.9	22.9	24.0	23.9	21.0	17.3	19.7		
		MCDBR	20.0	25.5	24.6	31.2	32.5	28.9	29.8	30.9	34.8	30.0	20.9	20.5		
		MCWBR	14.8	19.0	15.8	16.0	14.6	12.8	14.0	15.0	17.4	16.9	13.9	15.7		
		MCDBR	19.9	24.8	23.8	28.8	28.4	24.4	26.7	27.7	31.8	28.0	19.7	20.7		
	5% WB	MCWBR	15.3	18.8	15.7	15.5	13.1	11.8	13.6	14.4	16.2	16.6	13.9	16.3		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.266	0.313	0.327	0.343	0.353	0.355	0.362	0.307	0.281	0.254	0.240		
	taud		2.304	2.354	2.297	2.388	2.391	2.377	2.416	2.349	2.501	2.547	2.455	2.310		
	Ebn at Noon		231	270	276	287	287	284	281	272	278	264	237	216		
	Edh at Noon		18	24	32	33	35	36	34	34	26	20	16	15		
(44) All-Sky Solar Radiation	RadAvg		305	635	1125	1482	1817	1909	1951	1606	1108	636	304	225		
	RadStd		34	49	86	139	135	124	95	111	114	75	34	21		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page